



Physiology of Small Intestine

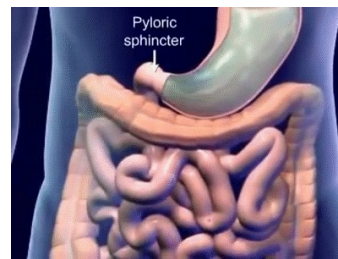
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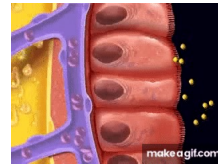
- **Introduction to small intestine.**
- **Regions of the small intestine and their functions.**
- **Types of movement that occur in the small intestine.**
- **Villi, Microvilli, and Circular folds.**
- **Brush border enzymes.**
- **Intestinal digestion and absorption of:**
 - **Carbohydrates, Proteins, and Lipids.**
 - **Electrolytes, Vitamins, and Water.**

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Objectives:

After studying this lecture, you will be able to:

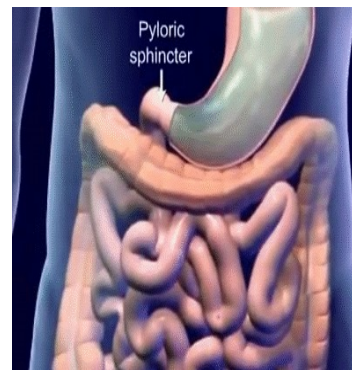
- Describe the small intestine.
- List the regions of the small intestine and their functions.
- Describe the importance of villi, microvilli, & circular folds.
- List the Brush border enzymes.
- Describe the intestinal digestion and absorption of:
 - Carbohydrates, Proteins, and Lipids.
 - Electrolytes, Vitamins, and Water.



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Introduction: The small intestine:

- It extends from the **pyloric sphincter** to the **ileocecal junction**.
- Its length is:
 - ~3 meters long (living person)
 - ~6.5 meters long (cadaver)
- Average diameter= **2.5 cm**.
- Is the principal site for nutrient **digestion** and **absorption**.

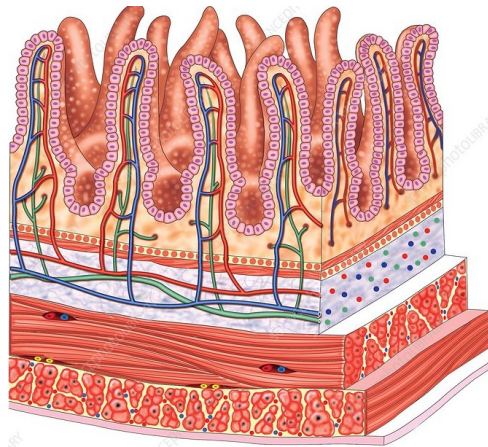


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Introduction: The small intestine:

- The wall of the small intestine has 4 layers:

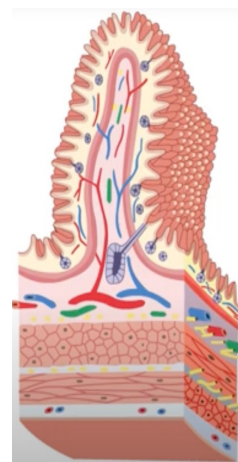
- **Mucosa,**
- **Submucosa,**
- **Muscularis,**
- **Serosa.**



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Introduction: The small intestine:

- The mucosa contains **villi** and **microvilli**, dramatically **increasing surface area**.
- The submucosa houses **Brunner's (duodenal) glands**, which produce **alkaline mucus**.
- The muscularis consists of **circular** and **longitudinal smooth muscle**, **essential for motility**.

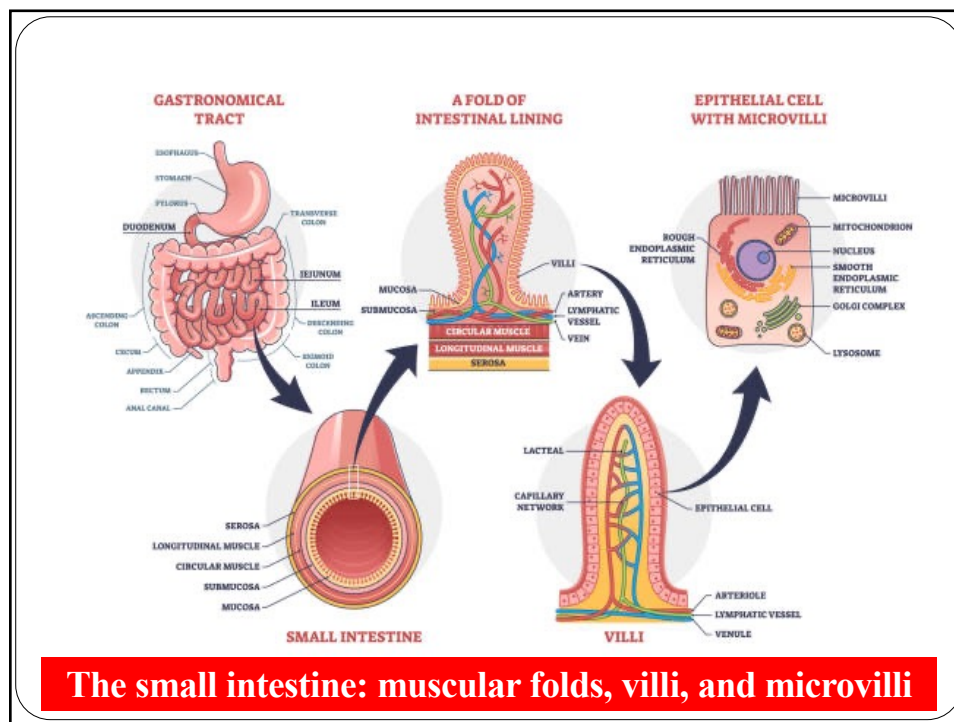


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Introduction: The small intestine:

- It is **functionally adapted** for efficient nutrient **breakdown, absorption, and motility**.
- It includes **folds, villi, and microvilli**, which together form an **extensive absorptive surface area**.
- **Capillaries and lacteals** within villi **transport absorbed nutrients into circulation**.

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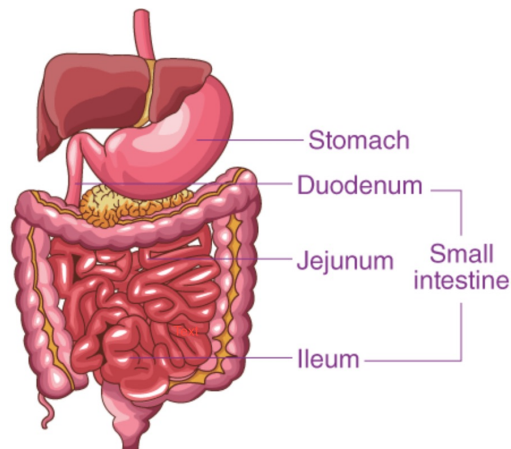
The small intestine: muscular folds, villi, and microvilli

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Regions of the Small Intestine:

The small intestine is divided into 3 regions:

- **Duodenum:**
- **Jejunum:**
- **Ileum:**



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Duodenum:

- The first and the **shortest region** (~25 cm long).
- It curves around the head of the pancreas.
- Plays a central role in receiving chyme and digestive secretions.
- It receives secretions from the **liver**, **gallbladder**, and **pancreas**, which aid digestion.

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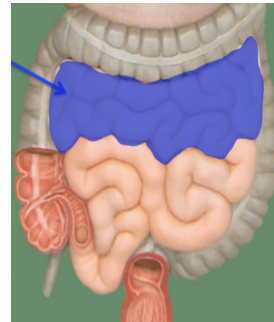
Duodenum:

- **Brunner's (Duodenal) glands:**
 - ❑ They are located in the duodenal submucosa and secrete **alkaline mucus**, that:
 - **neutralize acidic chyme** entering from the stomach.
 - **provide an optimal pH** for pancreatic enzyme activity.
 - **prevents damage** to the intestinal mucosa.
- Duodenum also contains endocrine cells that secrete **Secretin** and **Cholecystokinin**.

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Jejunum:

- It is located in the middle and is especially **specialized for absorption**.
- It accounts for **~2/5th** of the small intestine.
- It is **richly supplied with blood vessels** and has **prominent circular folds** (**plicae circulares**).



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Jejunum:

- Its mucosa contains abundant **villi** that **maximize nutrient absorption**.
- The jejunum is the main site for **absorption of**:
 - **Carbohydrates,**
 - **Proteins,**
 - **Lipids.**

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Ileum:

- It is the **longest portion** ~**3/5th** of the small intestine.
- Ileum has fewer folds and villi compared to the jejunum, but it still **contributes significantly to absorption**.

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Ileum:

- It is specialized for the uptake of **bile salts** and **vitamin B12**, which are essential for **digestion** and **hematopoiesis**.
- The ileum ends at the **ileocecal valve**, which **regulates the passage of contents into the large intestine**.

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Ileum: Peyer's patches:

- **Peyer's patches** are lymphoid aggregates found in the ileum.
- They provide **immune surveillance** against pathogens entering with food.
- They contain B and T lymphocytes, **supporting gut-associated immunity**.

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Intestinal Juice:

- a watery mixture containing mucus and enzymes secreted by the glands of the small intestine.
- It **provides the medium in which digestive processes occur.**
- The juice contains enzymes that complete the digestion of carbohydrates and proteins.
- The secretions also **lubricate the intestinal walls and facilitate transport of nutrients.**

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Brush Border Enzymes:

- The brush border contains enzymes such as **lactase, maltase, sucrase, and peptidases.**
- They are bound to the **microvilli membranes** rather than secreted into the lumen.
- **They allow final hydrolysis of:**
 - **disaccharides → monosaccharides**
 - **peptides → amino acids.**

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Motility Patterns of the Small Intestine:

- The small intestine exhibits **segmentation** and **peristalsis**.

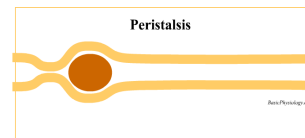
- **Segmentation:**

- involves local contractions that **mix** chyme with digestive secretions.



- **Peristalsis:**

- involves waves of contraction that **move chyme forward** and then **maximizes digestion & absorption**.



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Digestion of Carbohydrates:

- Carbohydrate digestion begins with salivary and pancreatic amylase.
- **Amylase** breaks down **polysaccharides** into **disaccharides**.
- Brush border enzymes convert **disaccharides** into **monosaccharides**.
- **Only monosaccharides** like **glucose**, **galactose**, and **fructose** are **absorbed**.

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Absorption of Carbohydrates:

- Glucose and galactose are absorbed via sodium-dependent **co-transport**.
- Fructose is absorbed by **facilitated diffusion**.
- Eventually, all monosaccharides enter the bloodstream.

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Digestion of Proteins:

- Proteins are initially digested in the stomach by pepsin.
- In the small intestine, pancreatic enzymes (**trypsin**, **chymotrypsin**, and **carboxypeptidase**) further **hydrolyze** proteins.
- Brush border **peptidases** complete the breakdown to **amino acids (AAs)**.
- Both **AAs** and **small peptides** can then be **absorbed**.

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Absorption of Proteins:

- **AAs** are absorbed by **sodium-dependent active transport systems**.
- **Dipeptides** and **tripeptides** are absorbed via proton-coupled peptide transporters (**PepT1**).
- Once inside the **enterocyte**, dipeptides and tripeptides are hydrolyzed into **AAs**.
- Absorbed **AAs** enter the blood and are transported to the liver.

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Digestion of Lipids:

- Lipids require **emulsification** by bile salts into **micelles**.
- **Pancreatic lipase** hydrolyzes **triglycerides** into **monoglycerides** and **free fatty acids**.
- **Phospholipase** and **cholesterol esterase** further process dietary fats.
- **Micelles** transport lipid products to the epithelial surface for absorption.

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Absorption of Lipids:

- Lipid products **diffuse** across the intestinal epithelium into **enterocytes**.
- Inside cells, they are re-esterified into triglycerides.
- **Triglycerides** combine with **cholesterol**, **phospholipids**, and **proteins** to form **chylomicrons**.
- **Chylomicrons** are secreted into **lacteals** and transported through the lymphatic system.

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Absorption of Vitamins:

- Fat-soluble vitamins (A, D, E, K) are absorbed with dietary lipids via micelles.
- Water-soluble vitamins use **diffusion** or **carrier-mediated mechanisms**.
- Vitamin B12 requires intrinsic factor for absorption in the **ileum**.
- This highlights the importance of intact intestinal mucosa for micronutrient uptake.

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Absorption of Minerals:

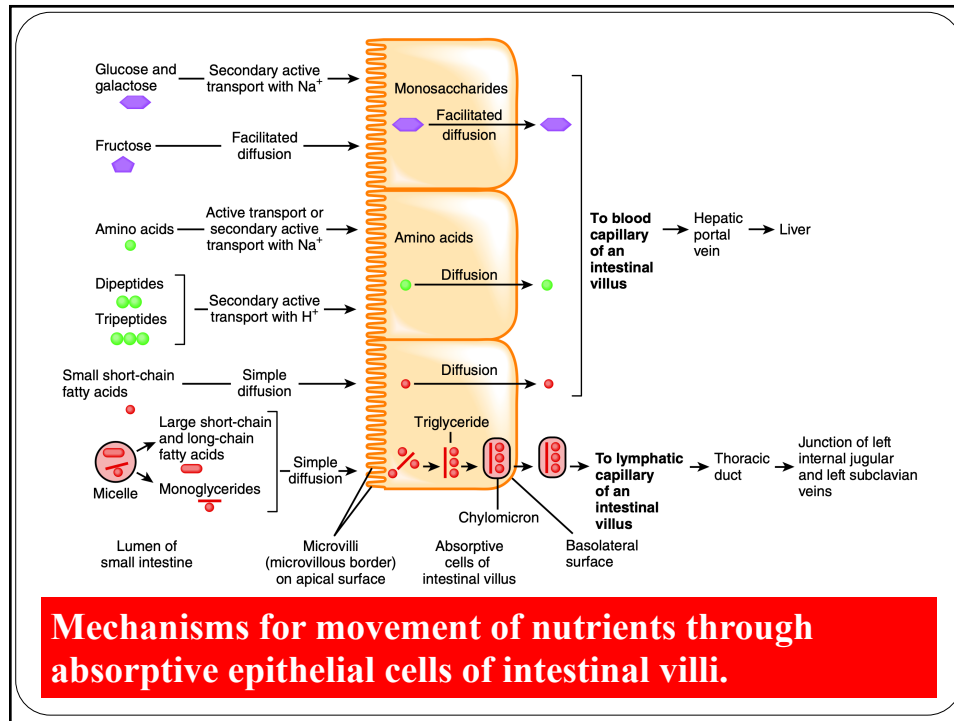
- **Calcium (Ca^+) absorption** occurs in the duodenum and **is vitamin D-dependent**.
- **Iron absorption** is **enhanced by gastric acid** and occurs mainly in the duodenum.
- **Magnesium, phosphate, and zinc** are absorbed **throughout the small intestine**.

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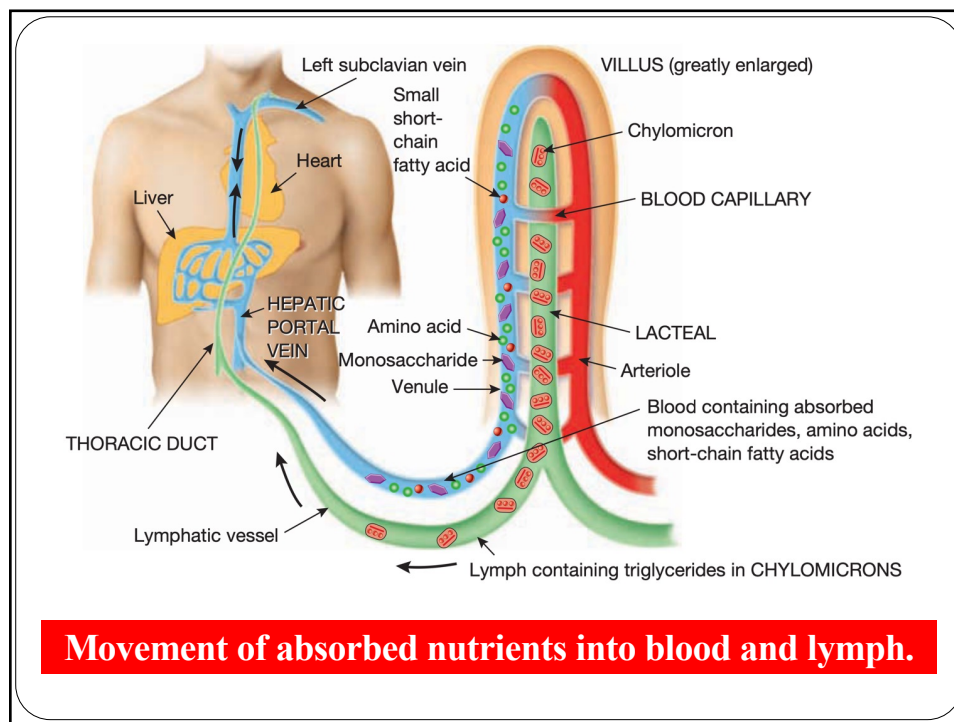
Water and Electrolytes:

- The small intestine absorbs large volumes of water and electrolytes daily.
- Na^+ absorption is coupled with nutrient absorption and also occurs via N^+ channels.
- Cl^- is absorbed passively or in exchange with HCO_3^-
- K^+ absorption occurs mainly by diffusion.

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Neural Control:

- The enteric nervous system regulates intestinal motility and secretion.
- **Parasympathetic stimulation** ↑↑ peristalsis & secretion
- **Sympathetic stimulation** ↓↓ motility & blood flow
- The **coordination** ensures optimal digestion and absorption under varying conditions.

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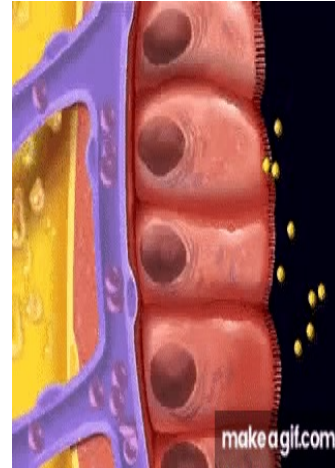
Immune Function:

- The small intestine is not only a digestive organ but also an immune barrier.
 - **Peyer's patches** and **intraepithelial lymphocytes** protect against pathogens.

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Functions of the Small Intestine:

1. Segmentations mix chyme with digestive juices and bring food into contact with mucosa for absorption; peristalsis propels chyme through small intestine.
2. Completes digestion of carbohydrates, proteins, and lipids; begins and completes digestion of nucleic acids.
3. Absorbs ~90% of nutrients and water that pass through digestive system.



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In Summary:

Lacteal

- One of many **lymphatic vessels** in villi of the intestines that **absorb triglycerides and other lipids from digested food**.

Duodenal glands

- Secrete alkaline fluid to buffer stomach acids and mucus for protection and lubrication.

Circular folds

- Folds of mucosa and submucosa that increase surface area for digestion and absorption.

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In Summary:

Intestinal villi:

- Fingerlike projections of mucosa that are sites of absorption of digested food and increase surface area for digestion and absorption.

Microvilli:

- Microscopic, membrane-covered projections of absorptive epithelial cells that contain microvillous border enzymes and increase surface area for digestion and absorption.

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In Summary:

■ Complete Digestion of Food

- The **partially digested food** is absorbed by the duodenum of the small intestine along with the digestive juices from the liver, pancreas and its own walls.
- The liver secretes the **bile juice**, which converts fat into tiny droplets so that their digestion becomes easy.

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In Summary:**■ Complete Digestion of Food**

- The pancreas produces **pancreatic juice** that breaks down fats into fatty acids and glycerol.
- The **intestinal juice** secreted by the walls of the small intestine:
 - ✓ breaks down carbohydrates into simple sugars.
 - ✓ converts the proteins into amino acids.
- All these simple, broken down forms are called the **digested food**.

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In Summary:**■ Absorption of Digested Food**

- The food that is digested is absorbed into the blood vessels in the walls of the intestine.
- The **finger-like projections (villi)**, drastically increase the surface area of the small intestine for greater absorption of the digested food.

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In Summary:

■ Absorption of Digested Food

- The blood carries the absorbed food material to different parts of the body.
- Glucose breaks down to form O_2 and CO_2 and releases the **energy required** for various life processes.
- The **undigested** and **unabsorbed food** passes from the small intestine to the **large intestine**.

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Questions/Comments



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MCQs

What is the main function of bile salts in the small intestine?

- A. Protein digestion
- B. Carbohydrate absorption
- C. Emulsification of fats
- D. Vitamin synthesis
- E. Acid neutralization

Which hormone stimulates pancreatic bicarbonate secretion?

- A. Secretin
- B. Gastrin
- C. Cholecystokinin
- D. Motilin
- E. Somatostatin

MCQs

Which of the following is absorbed via lymphatics in the small intestine?

- A. Glucose
- B. Amino acids
- C. Fatty acids and monoglycerides
- D. Water
- E. Electrolytes

The main site of iron absorption in the small intestine is:

- A. Sigmoid colon
- B. Jejunum
- C. Ileum
- D. Cecum
- E. Duodenum

MCQs

Which type of transport is used for glucose absorption in the small intestine?

- A. Simple diffusion
- B. Facilitated diffusion
- C. Active transport with sodium
- D. Endocytosis
- E. Osmosis

In the duodenum, the absorption of is vitamin D-dependent:

- A. Na^+
- B. Water
- C. Ca^+
- D. K^+
- E. Mg^+

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MCQs

The average length of small intestine in living persons is:

- A. ~2.0 meters
- B. ~2.5 meters
- C. ~3.0 meters
- D. ~5.5 meters
- E. ~6.5 meters

Which hormone stimulates gallbladder contraction and bile secretion?

- A. Secretin
- B. Gastrin
- C. Motilin
- D. Somatostatin
- E. Cholecystokinin

-MCQs Answers:

1- C

2- A

3- C

4- E

5- C

6- C

7- C

8- E